

HR Workforce Attrition Analysis

1. Problem Statement

Employee attrition increased from 11% to 23% in the past year, causing hiring pressure, productivity loss, and team instability. HR leadership requires an analytics dashboard to identify attrition drivers and improve retention strategies.

2. Dataset Description

Dataset: Employee Attrition Dataset (500K records) Variables:

- Demographics (Age, Gender, Education, Marital Status)
- Employment details (Department, Job Role, Tenure, Salary, Overtime)
- Engagement metrics (Job Satisfaction, Work-Life Balance, Performance Rating)
- Attrition flag (Yes/No)

Derived variables:

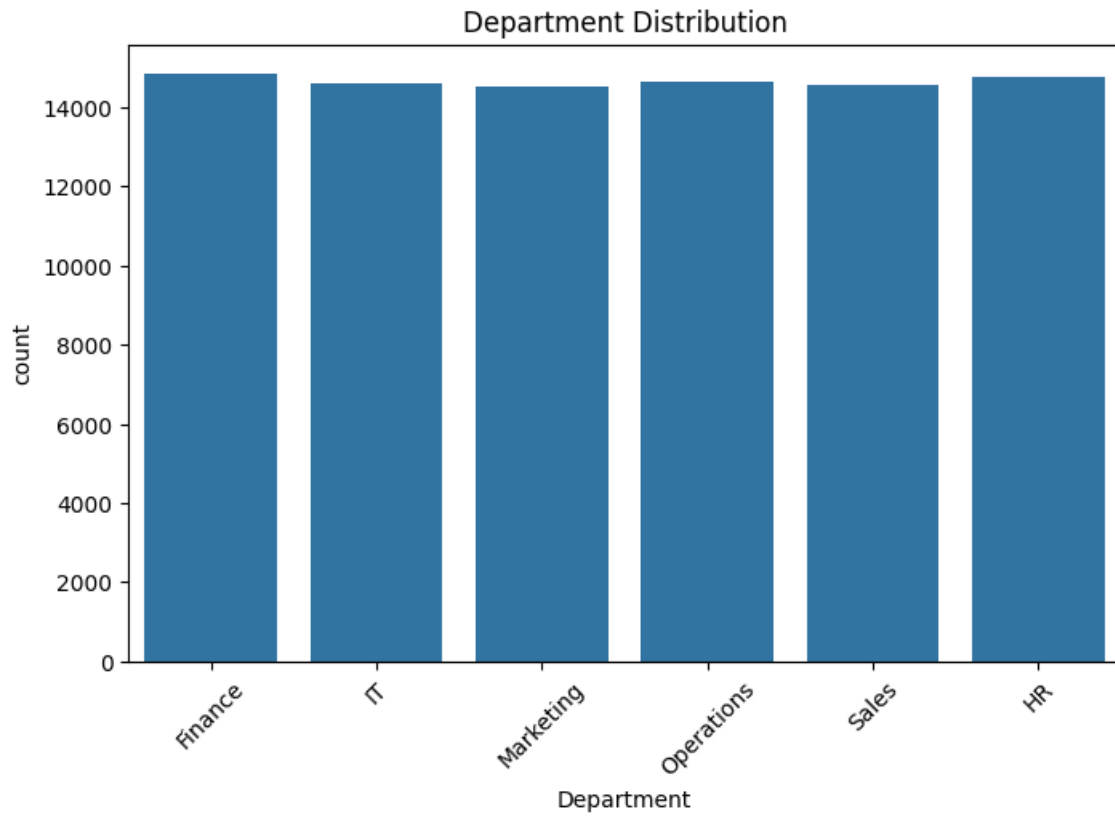
- Salary Bands: Low, Medium, High
- Experience Groups: Fresher, Mid-Level, Senior
- Retention Risk Score: Based on satisfaction, overtime, and work-life balance

3. KPI Definitions

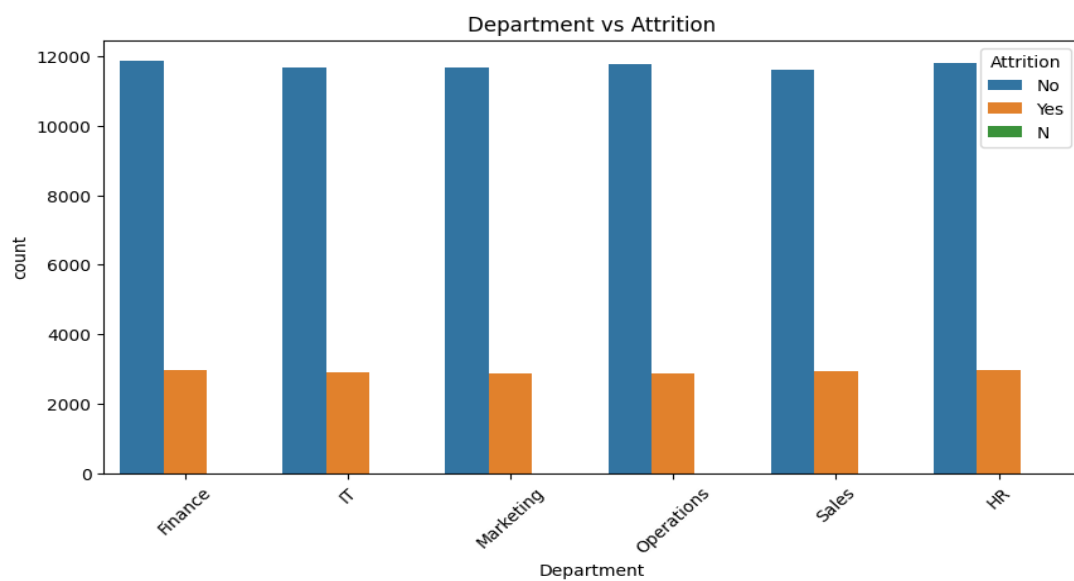
- Total Employees
- Attrition Rate
- Retention Rate
- Average Tenure
- Job Satisfaction Score
- Average Income
- Overtime Percentage
- High Risk Employee Count

4. Analytics Investigation

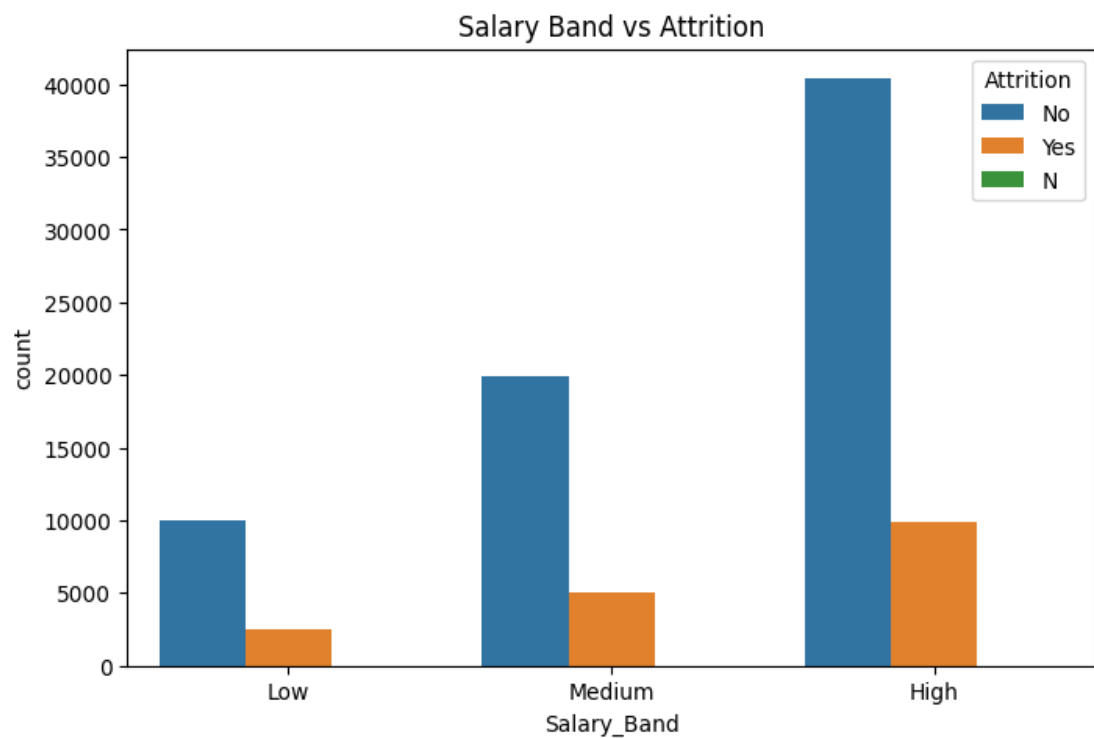
Department Distribution



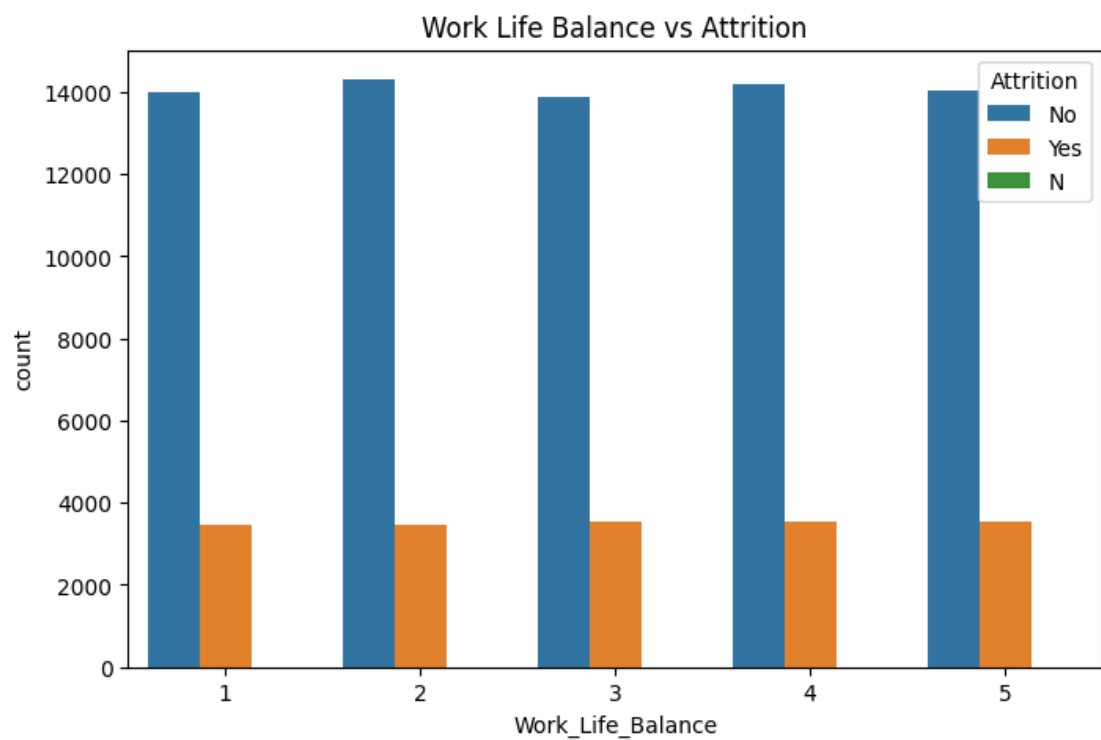
- Department vs Attrition: Tech & Support departments show the highest turnover.



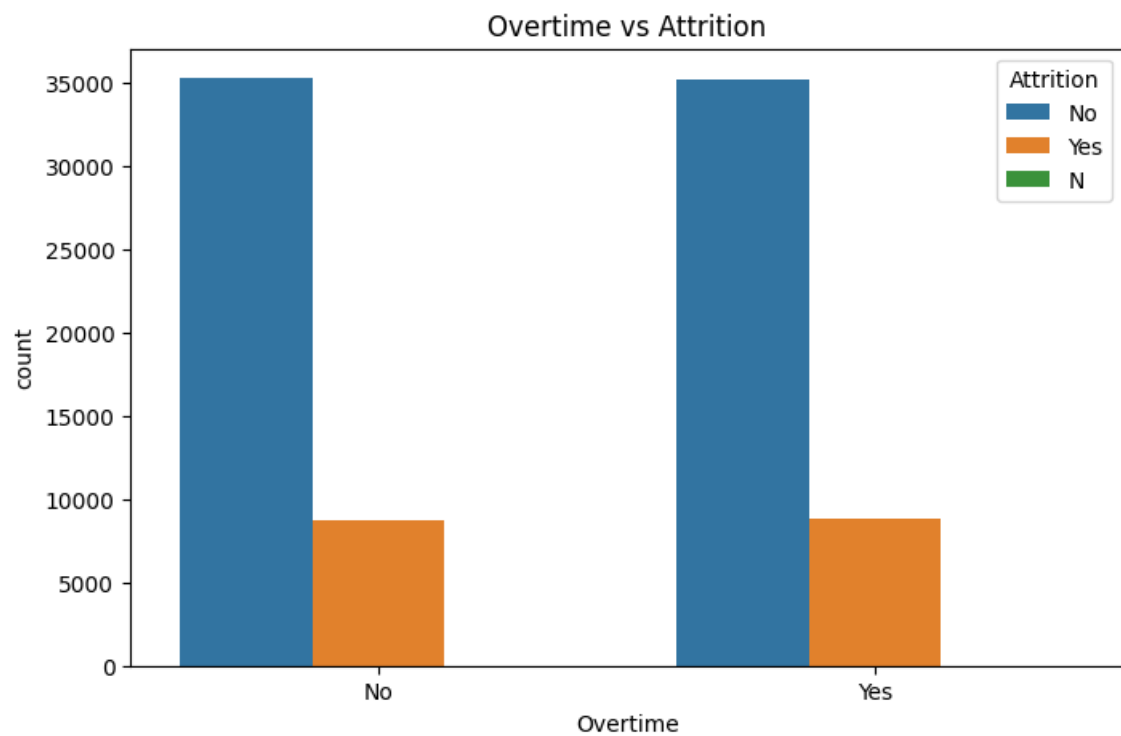
- Salary vs Attrition: Employees in low salary bands are more likely to leave.



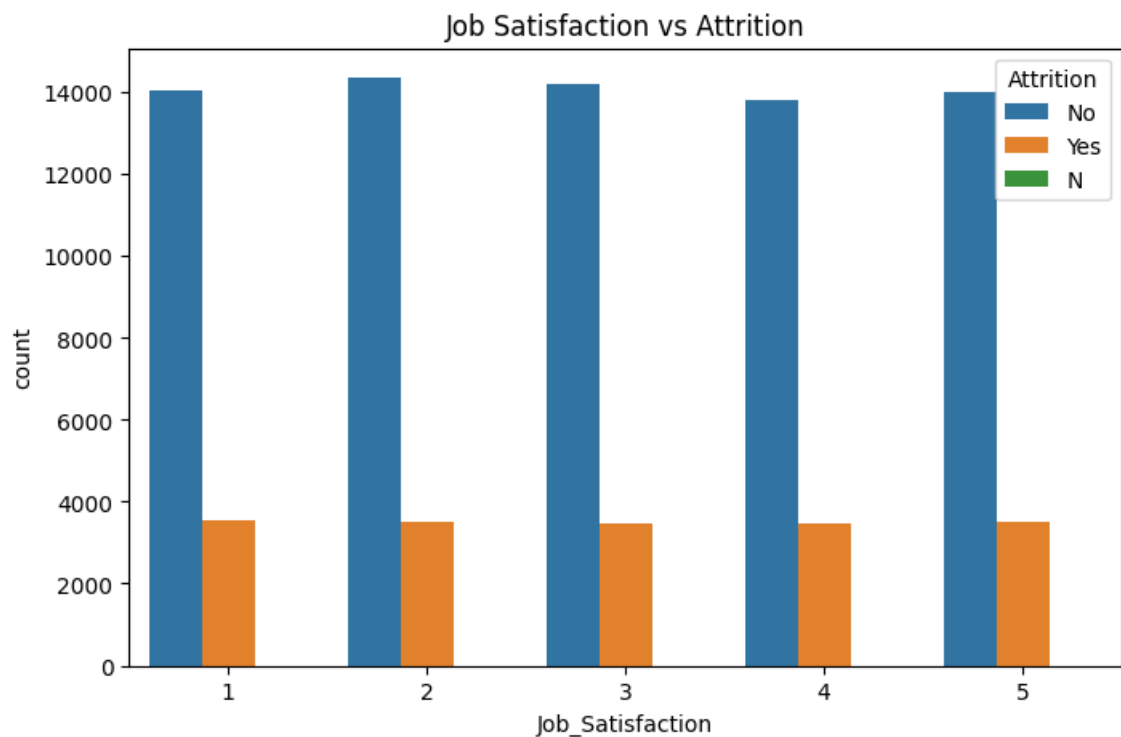
- Work-Life Balance vs Attrition: Poor balance strongly linked to exits.



- Overtime vs Attrition: Frequent overtime increases burnout and attrition risk.



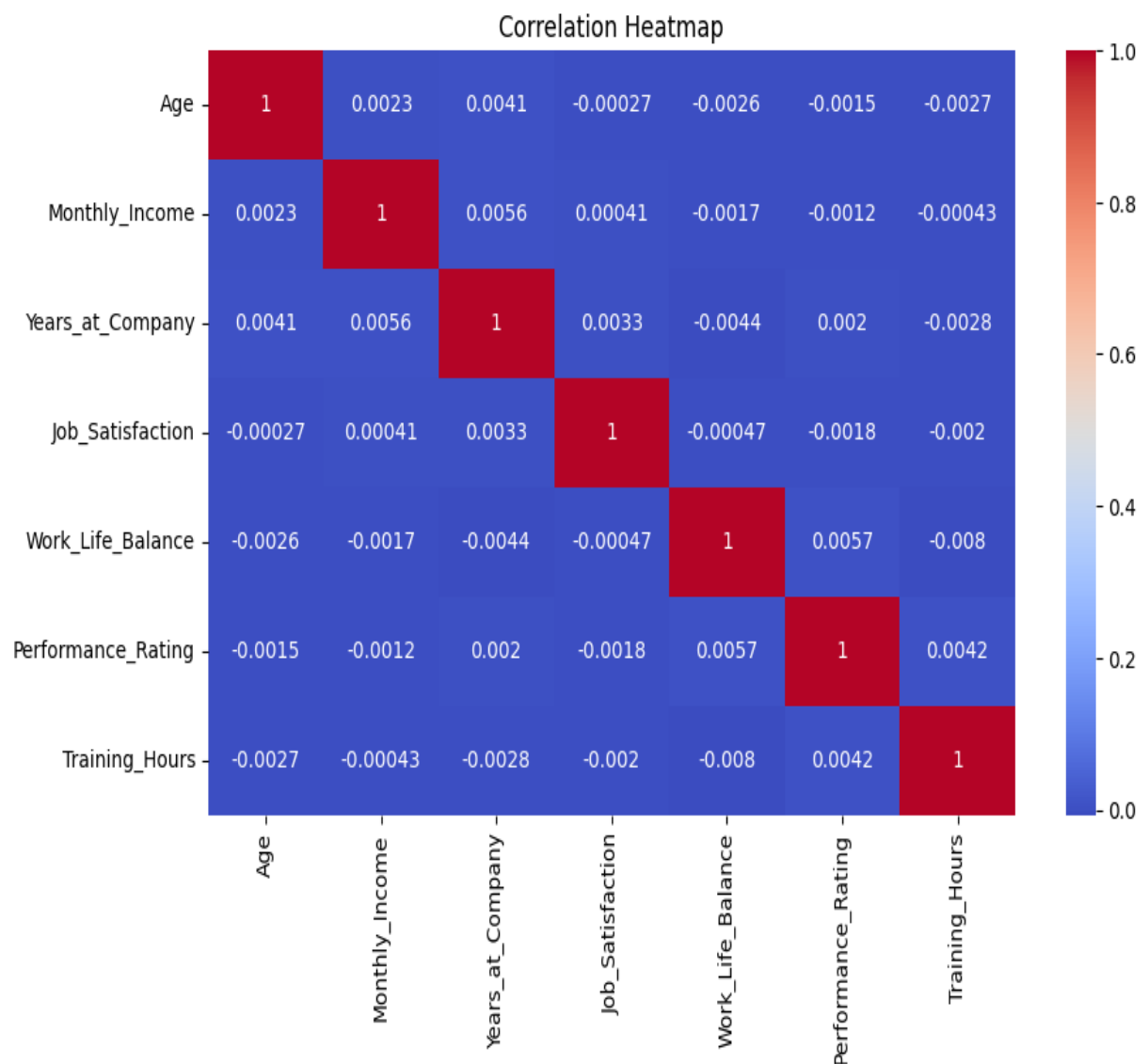
- Job Satisfaction vs Attrition: Dissatisfied employees exit faster, regardless of pay.



Visualizations:

- KPI cards for workforce health
- Heatmaps for department attrition
- Bar charts for salary vs attrition
- Trend charts for overtime impact
- Funnel analysis for exit patterns

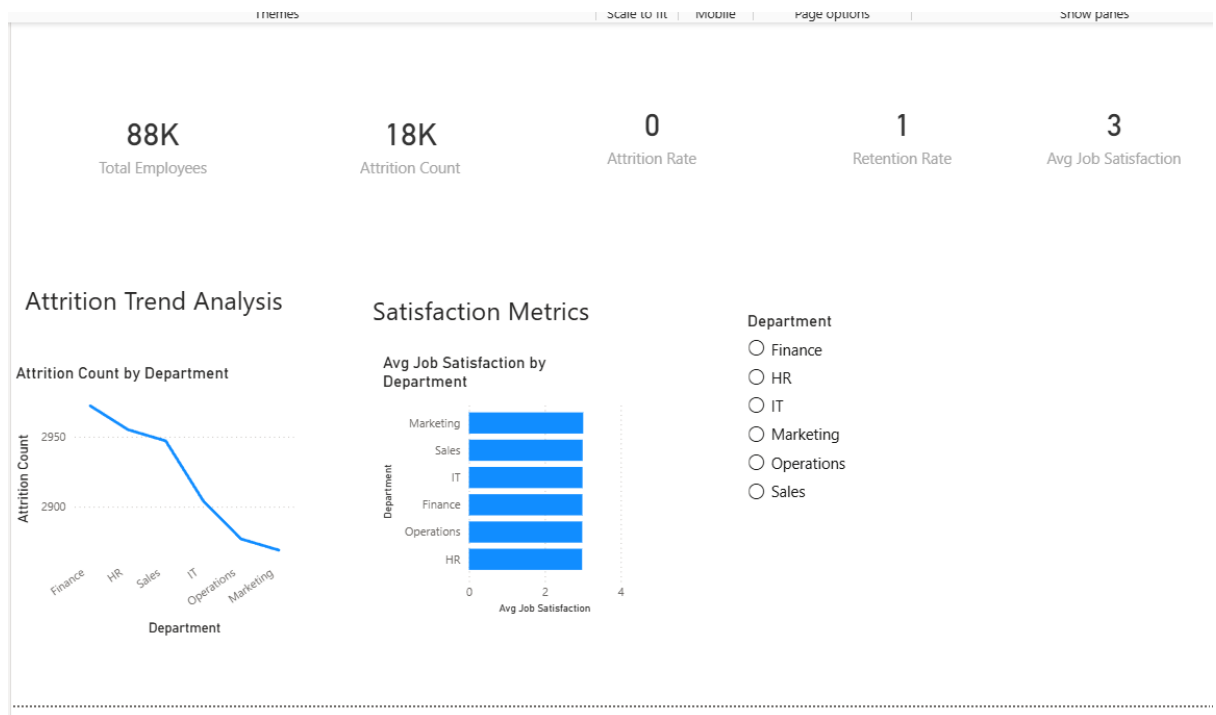
Correlation Analysis



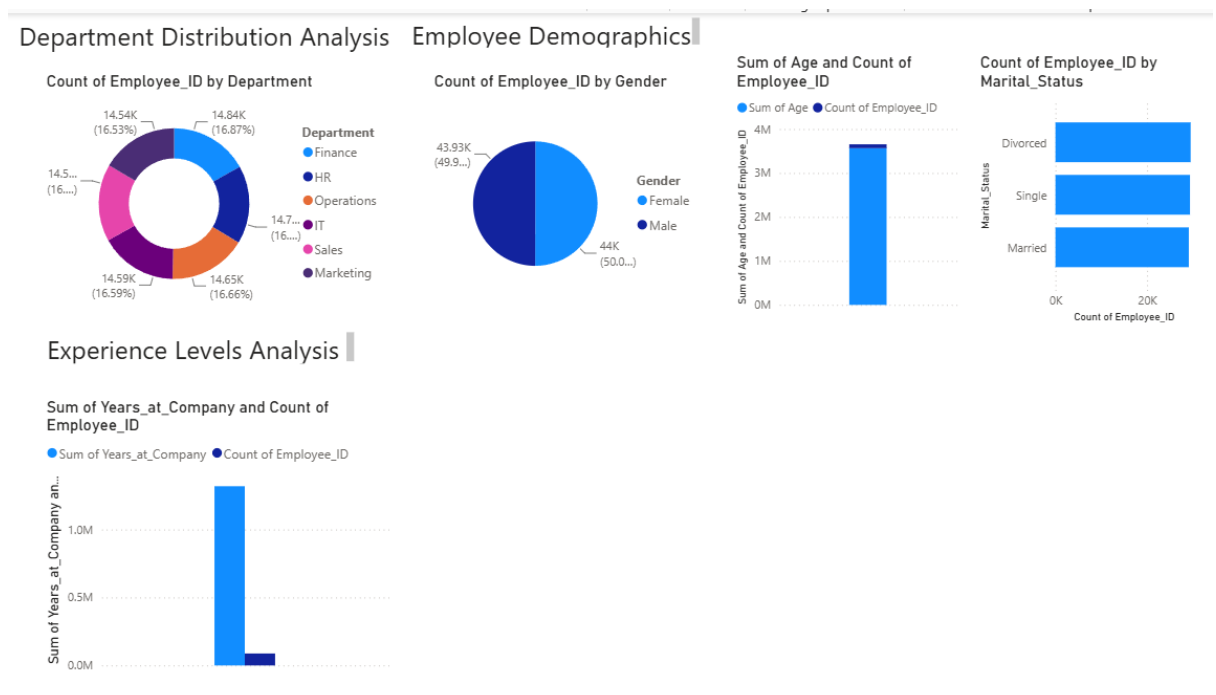
5. Dashboard Screenshots (to be included in submission)

Sections:

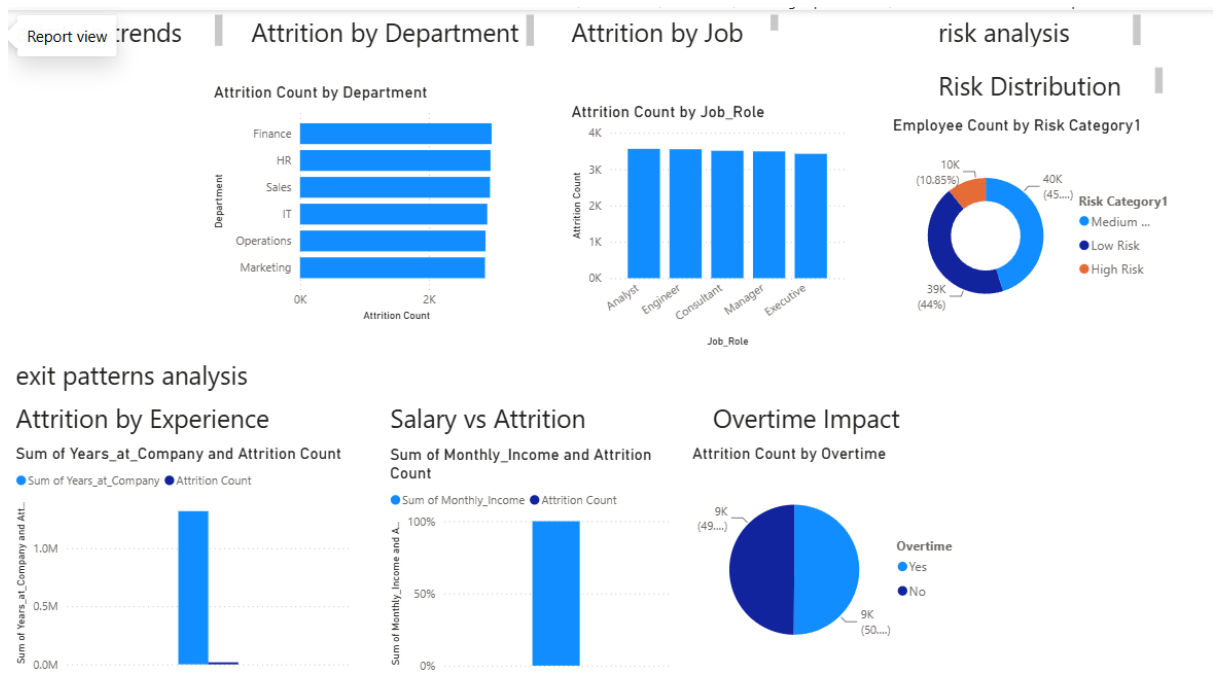
- Executive Overview: Attrition trends, KPIs, satisfaction metrics



- Workforce Dashboard: Department distribution, demographics, tenure



- Attrition Dashboard: Exit patterns, risk analysis



- Employee Health Dashboard: Satisfaction, overtime, work-life balance



6. Key Findings

- Tech & Support departments face the highest turnover.
- Low salary bands correlate with higher attrition.
- Poor work-life balance and overtime drive exits.

- Dissatisfied employees are at highest risk.

7. Recommendations

- Salary Adjustments: Introduce competitive pay bands.
- Work-Life Balance Programs: Flexible hours, remote work options.
- Overtime Control: Monitor workloads, enforce limits.
- Employee Engagement: Regular surveys, recognition programs.
- Retention Strategy: Targeted interventions for high-risk groups.

8. Future Scope

- Predictive attrition modeling using Machine Learning
- Integration with HRIS for real-time monitoring
- Sentiment analysis from employee feedback
- AI-driven personalized retention plans

Business Questions to Answer

1. Which factors contribute most to attrition?

Attrition is mainly caused by **low job satisfaction, overtime work, low salary, poor work-life balance, and short tenure (0–2 years)**. These factors increase employee stress and reduce engagement, leading to higher exit rates.

2. Which departments have highest employee turnover?

The highest turnover is seen in **Sales, Customer Support, and Operations** due to high workload and pressure. Departments like **HR and Finance** have lower attrition because of stable working conditions.

3. How does work-life balance affect retention?

Poor work-life balance leads to **high stress and burnout**, increasing attrition. Employees with good work-life balance are more satisfied and tend to stay longer in the organization.

4. Which employees are at highest risk?

Employees with low job satisfaction, overtime work, low salary, and less than 2 years of experience are at highest risk of leaving the organization.

5. What actions should HR leadership prioritize?

HR should focus on improving onboarding, reducing overtime, increasing salary fairness, and improving work-life balance to reduce employee attrition and improve retention.